

Jinpeng Lu

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Research Profile

I work on controllable video generation and seek core foundation-model R&D in pre-training, RL post-training or research. At X-Humanoid, I developed camera conditioning and WRBench, and explored action-conditioned world modeling and few-step action prediction. Publications include first-author papers at ICLR 2026 and MICCAI 2024 and collaborative work at EMNLP 2026 and MIA 2025, with 163 Google Scholar citations and an h-index of 6 overall.

Education

University of Science and Technology of China

M.S. in Information and Communication Engineering

Hefei

Sep. 2025 to Jun. 2028, expected

Harbin Institute of Technology, Shenzhen

B.S. in Data Science and Big Data Technology; GPA 3.728/4, rank 17/172, top 10%

Shenzhen

Sep. 2021 to Jun. 2025

Internship Experience

Beijing Innovation Center of Humanoid Robotics X-Humanoid, UBTECH

Beijing

Embodied World Model Team, Large Model Algorithm Intern

Nov. 2025 to Oct. 2026

- **Controllable video generation and world-state evaluation:** Independently led camera-control research, from pose conditioning and multi-source data pipelines to training and evaluation. Diagnosed reliance on motion text and failures to preserve object dynamics. Proposed WRBench as first author to test event persistence under viewpoint changes, delivering benchmark design, automatic scoring, human pairwise evaluation, and the paper.
- **Exploratory embodied world modeling:** Tested few-step action-policy sampling and fixed an initialization defect that disabled action conditioning; conducted preliminary controlled checks.
- **Collaborative research:** Contributed to the core design and refinement of VLA-Trace, accepted to EMNLP 2026; handled data processing for Pelican-Unify 1.0, supporting joint generation of future video and actions.

Research Experience

Controllable Video Generation and Spatiotemporal Consistency

Multi-Shot Video Storytelling: Cross-Shot Consistency and Event Evolution

May to Sep. 2026

- **Task and data pipeline:** Built a 184K-story training set for consistent characters and script-driven events. Aligned shots and captions across sources; retained separate quality scores and sample weights for visual quality, text coverage, and content risk. Excluded held-out samples to support controlled data and model comparisons.
- **Model design and iteration:** Designed joint flow matching of story and entity representations and video latents on Wan2.2-TI2V-5B; explored clean keyframe prefixes with post-event captions for sequential generation. Ablated semantic conditions and memory, and replaced references to distinguish effects on shot structure, appearance, and state updates.
- **Evaluation and diagnosis:** Evaluated identity, shot transitions, relations, and event outcomes against scripts. Validated judges with synthetic failures and story-level bootstrap intervals. Found that appearance metrics reward repetition and generated memory matching prior-shot text can induce copying, exposing tension between identity and new-shot instructions. Method gains remain under investigation.

WRBench: World-State Evaluation and Diagnosis under Viewpoint Changes

Feb. to Jul. 2026

- **Benchmark and data design:** Tested event persistence beyond camera control and visual quality by moving the camera away and back while independently varying relocation and state change. Built scenes, first frames, and generation prompts; unified pose, action, geometry, and language controls across 25 scene families, 23 models, and 9,600 videos.
- **Scoring and human validation:** Combined camera-trajectory estimation, visual features, and vision-language questions to separate camera execution, visual integrity, and visible and reobserved consistency. Used answer probabilities from paired positive and negative questions to reduce scoring bias; built a web annotation tool and organized blind pairwise evaluation, with evaluator-to-human rank correlation of 0.54 to 0.71.
- **Findings:** Camera direction affected re-observation availability, which did not ensure correct target positions or states. Scale and release gains in the evaluated Wan-Fun series mainly improved visible consistency. Released an arXiv preprint as first author, identifying a gap between camera control and persistent event-state modeling.

- **Data processing:** Integrated natural and embodied video sources, estimated missing poses with ViPE, and normalized coordinates, trajectory scales, and frame rates. Implemented trajectory filtering and motion-based sampling; removed camera-motion text to reduce shortcuts that bypass pose conditioning.
- **Conditioning and training:** Implemented per-pixel Plücker-ray and relative-pose conditioning on Wan2.2-TI2V-5B through input-level and layer-wise injection. Adapted ReCamMaster-style pose encoding to image-to-video generation; studied frozen-backbone training, full fine-tuning, and mixed-source data for trajectory following and dynamics preservation.
- **Experimental diagnosis:** In embodied scenes, synthetic-data training enabled camera following but struggled to preserve arm motion; real embodied data improved dynamics while weakening control. Used text/pose controls and trajectory visualization to analyze failures, motivating WRBench's study of coupled camera motion and event evolution.

Efficient Visual Representation and Multimodal Fusion

VeloxSeg: Efficient 3D Medical Segmentation and Knowledge Transfer

Ended ~Nov. 2025

- **Efficient architecture:** Addressed detail loss and costly multimodal interaction with Johnson-Lindenstrauss-guided convolution group sizes and paired-window attention for parallel multi-scale local and global modeling. Combined convolution and attention in a 1.66M-parameter multimodal medical segmentation network.
- **Transfer and ablation:** Matched feature-channel Gram matrices to transfer texture from self-supervised reconstruction without spatially aligning tasks with different regions of interest. Ablated architecture, group sizes, and distillation choices to analyze accuracy and compute; the teacher adds no inference overhead. Accepted to ICLR 2026 as a first-author paper.

Selected Publications: First and Co-first Author

ICLR 2026

CCF-A

Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation

Jinpeng Lu, Linghan Cai, Yinda Chen, Guo Tang, Songhan Jiang, Haoyuan Shi, Zhiwei Xiong

[1 citation] Investigated how compact 3D models can preserve representation quality and fine detail. Combined JL-guided convolution with multiscale multimodal attention, then transferred reconstruction texture priors through Gram matrices to avoid spatial alignment conflicts across tasks. Distillation improved AutoPET-II Dice by 2.80 points at the same 1.66M parameters.

arXiv 2026

Current World Models Lack a Persistent State Core

Jinpeng Lu, Dexu Zhu, Haoyuan Shi, Linghan Cai, Guo Tang, Yinda Chen, Jie Cao, Duyu Tang, Yi Zhang, Yong Dai, Xiaozhu Ju

[7 citations, HF Daily Paper: 18 votes] Proposed WRBench to test whether rendering is decoupled from world-state evolution. Held events fixed while intervening on camera observations, separating re-observation evidence from state correctness. Across 23 models, viewpoint changes disrupted event evolution; improved visuals and control did not reliably yield persistent state.

arXiv 2025

Does DINOv3 Set a New Medical Vision Standard?

Che Liu*, Yinda Chen*, Haoyuan Shi*, Jinpeng Lu*, et al.

[54 citations, HF Daily Paper: 40 votes] Tested natural-image pretraining as a general medical representation by adapting frozen DINOv3 features to 2D/3D classification and 3D segmentation across model sizes and resolutions. It surpassed medical-specific models on some tasks, but scaling gains varied and transfer remained limited in pathology, electron microscopy, and PET, informing encoder selection and domain adaptation.

MICCAI 2024

CCF-B, top 11%

H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images

Jinpeng Lu, Jingyun Chen, Linghan Cai, Songhan Jiang, Yongbing Zhang

[18 citations] Addressed limited exchange between heterogeneous encoders by linking PET semantic localization and CT structural detail through local and global bidirectional attention, then selecting modality features by target relevance. Outperformed compared methods on AutoPET-II and Hecker2022; ablations and weight analysis supported hierarchical interaction and adaptive feature selection.

Service and Skills

Model Development

Controllable and multi-shot video generation: diffusion, flow matching, pose conditioning, and reference-frame memory; multimodal attention, self-supervised representation transfer, and knowledge distillation.

Research and Evaluation

Translate research questions into controlled experiments and diagnostic benchmarks; conditioning ablations, failure analysis, VLM scoring, human blind evaluation, and confidence intervals to assess gains and metric validity.

Data and Training

Multi-source video filtering, shot segmentation, caption alignment, and camera processing; Python, PyTorch, DeepSpeed ZeRO, distributed full fine-tuning, and training debugging.

Academic Service

Reviewer for ICLR 2027, NeurIPS 2027, and IEEE TNNLS.